

MATTHEW H. ZHANG

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EDUCATION

The University of Texas at Austin – GPA: 3.52 May 2028
Bachelor of Science in Electrical and Computer Engineering Honors
Bachelor of Business Administration in Canfield Business Honors
Relevant Coursework: Circuit Analysis & Design Honors, Embedded Systems Honors, Computing Systems Honors, Software Design & Implementation Honors, Data Science Honors

EXPERIENCE

Hardware Engineer Intern – eve Vehicles Corporation May 2026 – Present
• Designing battery quick-swap and power distribution board with integrated sensing for autonomous UAVs

Embedded Systems Engineer – GHOST Robotics Nov 2025 – Present
• Validated custom sensor PCBs and integrated them into tight mechanical assemblies using Onshape
• Redesigned differential I2C sensor hub PCB with address translation and configurable routing, reducing board area by 40%
• Performed board bring-up and debug by validating power rails and signals with a multimeter and logic analyzer
• Wrote interrupt-based RP2040 sensor polling firmware in C with custom sensor drivers over USB serial

Control Systems & Electrical Engineer – Texas Aerial Robotics Jan 2026 – Present
• Modeled inverted pendulum and TVC system dynamics and implemented PID, MPC, & LQR controllers in Python
• Designing compact AM32-based 30A 4S BLDC motor driver PCB under size and thermal constraints in Altium

Venture Analyst Intern – Tess Ventures Mar 2026 – Present
• Prepared technical demos and product-facing presentations for engineering and business audiences
• Evaluated robotics startups and provided technical due diligence for investment decisions

Team Captain & Lead Engineer – BagelBytes Robotics FRC #9400 Dec 2024 – Aug 2025
• Organized a team of 30+ individuals to successfully design, manufacture, and field functioning 125 lb robots
• Mentored 20+ members in CAD & electrical systems; improved team ranking by 1,500+ places during leadership
• Engineered custom CAN bus layout and architecture, reducing match-time electrical failures by 28%
• Designed 100+ custom sheet metal parts and 7 custom-built gearboxes with Onshape

PROJECTS

Inhabit, Open Source Universal Humanoid Teleoperation Rig Dec 2025 – Present
• Led development of a wearable motion capture system streaming human joint angles at 50 Hz over USB serial
• Designed MCU-based magnetic encoder PCB using RS-485 daisy-chaining, enabling plug-and-play joint modules
• Deployed and validated on Unitree G1 EDU for episodic task collection; redesigning v2 for robustness and UX

1st Place – Founders, Inc. World Tour Physical AI Hackathon May 2026
• Implemented classical, VLM, VLA, and IK control methods to autonomously play chess with a robot arm
• Utilized Docker containers and integrated with Sseed Studio's reBot B601-DM arm for piece manipulation
• Introduced CNN segmentation and end effector adjustments to increase pick and place accuracy by 61%

1st Place – Genesis x Texas Momentum x Yconic Hackathon April 2026
• Built a MCU-based wearable AI assistant with face recognition, live STT, and persistent memory of conversations
• Created a pipeline for face tracking, identity matching, and context retrieval across repeated interactions

Vinyl Record Player, Personal Project Dec 2025 – Present
• Independently designed MCU-based motor driver 2-layer PCB with USB-C PD power negotiation for turntable
• Created power distribution with custom buck converter (12V→5V), local decoupling, and differential-pair routing

SKILLS

Technical Skills: Circuit Design, PCB Layout, Firmware Development, Soldering, Prototyping, Control Systems, Computer Vision, CAD, Git, Debugging | **Tools:** KiCad, Altium, LTSpice, Oscilloscope, Logic Analyzer, Linux, Onshape, GDB, Docker | **Languages:** C, C++, Python, Assembly, R, SQL | **Embedded:** STM32Cube, ESP-IDF, CCSTUDIO, Ethernet, CAN, I2C, SPI, UART, ADC/DAC, BLE

AWARDS

Founders, Inc. Physical AI Hackathon 1st Place (2026) • UT x CSBA x HBA Hackathon 1st Place Track (2026) • Genesis x Texas Momentum x Yconic Hackathon 1st Place (2026) • IEEE RAS Robotathon 1st Place, Software Design Award (2025) • FIRST Robotics Rising All Stars Award (2025) • World's Rookie Inspiration Award, FIRST Robotics Worlds Qualifier (2024)